

ECON 5200: Research Proposal

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Causal Question

Does participation in the National Supported Work (NSW) job training program cause higher post-program earnings for disadvantaged workers?

Identification Strategy and Key Assumption

Method: Double Machine Learning (DML), using GradientBoostingClassifier and RandomForestClassifier as nuisance models for the propensity score, and GradientBoostingRegressor and RandomForestRegressor for the outcome model. Treatment is binary, so `discrete_treatment=True` is set to ensure proper propensity score estimation within [0, 1].

Key Assumption: Conditional Independence (Unconfoundedness) — conditional on observed covariates (age, education, race, marital status, pre-treatment earnings in 1974 and 1975), treatment assignment is independent of potential outcomes.

Why Prediction Alone Is Insufficient

A Random Forest achieves $R^2 = 0.421$ for predicting 1978 earnings, yet assigns `treat` a feature importance of only 0.0025 — nearly zero. This is because selection bias causes the predictive model to associate `treat=1` with *lower* earnings (the treated group's baseline is much lower). Prediction conflates treatment effects with pre-existing group differences. A causal method is required to isolate the true treatment effect after removing confounding.

Dataset

Source	Lalonde (1986) / Dehejia-Wahba (1999)
Construction	NSW treated (N=185) + CPS-1 comparison (N=15,992)
Total N	16,177 observations; 0 missing values
Treatment	<code>treat</code> (1 = NSW participant, 0 = CPS control)
Outcome	<code>re78</code> (real earnings in 1978, post-program)
Controls	<code>age</code> , <code>educ</code> , <code>black</code> , <code>hisp</code> , <code>marr</code> , <code>nodegree</code> , <code>re74</code> , <code>re75</code>
Benchmark	NSW RCT experimental ATE: ~\$1,794 (ground truth)

Preliminary Results

Naive OLS (no controls): -\$8,498 (95% CI: [-\$9,893, -\$7,102]). This suggests training *reduces* earnings — the opposite of the truth. The bias of -\$10,292 is caused by severe negative selection: NSW participants were chosen precisely because they were economically disadvantaged (mean pre-treatment earnings \$2,096 vs. CPS \$14,017). **Direction of bias:** downward, because negative selection into treatment underestimates the true positive effect.

DML (GBR nuisance): \$364 (95% CI: [-\$814, \$1,543]). **DML (RF nuisance, robustness):** \$1,541 (95% CI: [\$229, \$2,853]). Both specifications yield positive point estimates, correcting the naive bias. The RF estimate closely matches the experimental RCT benchmark of \$1,794, and its CI excludes zero. The overlapping CI region (\$229 to \$1,543) across both specifications supports directional robustness.

Primary Threat to Identification

Unobserved confounding: NSW participants were referred through probation officers and social workers based on characteristics not in the data (motivation, mental health, criminal history). If these factors also affect earnings, the conditional independence assumption fails. The RCT benchmark provides partial reassurance, but the observational estimate should be interpreted as conditional on observed covariates being sufficient to remove confounding.